ArchiMate Modelling Assessment – Case Study (Step 2)

Case Study Background – ClimateIQ - Weather Aggregation Platform Modernization

Industry Context

ClimateIQ is an environmental intelligence company delivering real-time weather insights to sectors like agriculture, logistics, renewable energy, and urban planning. Clients rely on timely and accurate weather information to optimize decision-making, reduce risk, and enhance operational efficiency.

ClimateIQ currently aggregates weather data from multiple sources:

- Government and commercial APIs (e.g., NOAA, OpenWeather, ECMWF)
- IoT sensors and ground stations
- Satellite and radar feeds

However, the system architecture is:

- **Manually integrated**, using scripts and point-to-point APIs
- **Not scalable** for new clients and higher data volume
- **Difficult to monitor and maintain**
- **Inconsistent in data quality** and real-time delivery

ClimateIQ seeks to overhaul this landscape to create a modular, scalable, and service-oriented Weather Aggregation Platform (WAP).

Business Challenge

ClimateIQ's customer base has grown rapidly over the past 18 months, especially in the agriculture and logistics sectors. However, its legacy architecture cannot handle increased data volumes or meet new demands for sub-hourly updates, regional granularity, and predictive analytics. Downtime and data inconsistency have already led to three major SLA breaches with key enterprise clients.

Additionally, integration complexity and lack of standard APIs are creating friction in onboarding new customers. The organization faces pressure from competitors offering more seamless,

cloud-native weather APIs with better developer experience. Without modernization, ClimateIQ risks losing key accounts and stalling further expansion.

Business Drivers

- Expand into global markets with scalable weather APIs
- Meet SLAs for real-time data delivery to enterprise clients
- Enable weather-based automation for smart agriculture and logistics
- Provide customizable weather alerts and predictive dashboards
- Integrate with client systems via developer-friendly, secure APIs
- Reduce operational costs by consolidating and containerizing legacy workloads
- Enhance brand reputation through data quality and service reliability

Transformation Objectives

ClimateIQ's Weather Aggregator Solution will:

- Ingest structured and unstructured weather data from over 50 diverse sources
- Normalize, enrich, and validate incoming data with location and time accuracy
- Feed insights into:
  - Interactive dashboards with real-time refresh
  - Customer-facing mobile applications
  - APIs and webhooks for enterprise system integration
- Support advanced analytics, ML-based prediction, and AI-based alerting
- Deploy on a cloud-native, event-driven microservices architecture
- Use container orchestration and CI/CD pipelines for continuous delivery
- Meet global data residency and security compliance standards (e.g., GDPR, ISO 27001)

Requirements

Functional Requirements

Ref	Category	As a...	Role	Need the ability to	Narrative	Priority
FR1	Functional	As a	Data Integration Engineer	Ingest weather data from multiple sources	Enable centralized ingestion of structured and	Must

				including APIs, satellite feeds, and IoT sensors	unstructured weather data from 50+ global sources	
FR2	Functional	As a	Data Processing System	Normalize, enrich, and validate incoming data	Ensure data quality, geospatial tagging, and timestamp accuracy for consistent downstream usage	Must
FR3	Functional	As a	Application Developer	Access weather data through secure, documented APIs	Provide REST and WebSocket APIs with OpenAPI specs to allow easy integration into client systems	Must
FR4	Functional	As a	Analytics Consumer	Receive AI-based weather predictions and alerts	Deliver short-term and long-term forecasts with confidence intervals for agriculture and	Should

					logistics optimization	
FR5	Functional	As a	Business User	View weather intelligence via dashboard	Provide an interactive web portal with customizable views, maps, and alert configurations	Should
FR6	Functional	As a	Integration Partner	Subscribe to weather events and alerts	Support event-driven architecture with message queue or webhook delivery for real-time notifications	Could
FR7	Functional	As a	Operations Manager	Monitor platform performance	Provide operational dashboards and alerts for latency, uptime, and error tracking	Must

Non-Functional Requirements

NFR1	Non-functional	As a	CTO	Ensure the platform	Implement elastic	Must

				scales to handle peak loads	scaling to support spikes in ingestion and delivery traffic	
NFR2	Non-functional	As a	Compliance Officer	Ensure compliance with data privacy and residency laws	Align with GDPR, ISO 27001, and relevant local regulations	Must
NFR3	Non-functional	As a	End User	Access services with high availability	Maintain uptime SLA of 99.95%	Must
NFR4	Non-functional	As a	Security Manager	Protect data integrity and confidentiality	Implement OAuth2, TLS encryption, and role-based access control	Must
NFR5	Non-functional	As a	Platform Engineer	Deploy and manage the platform efficiently	Use container orchestration and CI/CD pipelines to streamline deployment	Should
NFR6	Non-functional	As a	API Consumer	Experience low-latency responses	Ensure API response time under	Should

					200ms for common queries	

Key Stakeholders

Stakeholder	Role / Concern
Chief Technology Officer (CTO)	Ensure alignment with modernization roadmap and scalability targets
Product Managers	Define weather products, alerts, and APIs to match user expectations
Enterprise Clients	Agriculture and logistics firms who rely on up-to-date, actionable forecasts
Platform Engineers	Responsible for system stability, observability, and deployments
Data Scientists	Consume normalized weather feeds for analytics and AI modeling
Developers / Integration Partners	Use APIs and SDKs to embed ClimateIQ weather data into external systems

Architecture Vision

The new platform will embody the principles of **modularity**, **scalability**, and **automation**. Architecturally, the solution will feature:

- A clean separation between ingestion, processing, and delivery layers
- Clearly defined service boundaries and contracts via OpenAPI specifications
- Event streaming backbone (e.g., Kafka) for decoupled integration
- Elastic infrastructure powered by Kubernetes
- API Gateway and OAuth2 for secure client access
- Reusable application and infrastructure services aligned to business capabilities

This architecture aims to support ClimateIQ's vision of becoming the go-to platform for real-time, intelligent weather data globally.

Your Task (Candidate Instructions)

Model the **target architecture** for ClimateIQ's Weather Aggregator Platform using **ArchiMate 3.2**. The model should show how business drivers are supported by strategy, how capabilities are realized, and how application and infrastructure components work together to deliver weather services.

Self-Assessment Checklist

Before submitting, please validate your work using the following checklist:

- The use case is well-defined and scoped.
- At least 6 views are created which includes contextual, conceptual plus 4 others.
  - One must be a Technology view
  - Two cross-role or cross-layer views (e.g., Capability to Application, Strategy to Technology)
- Appropriate ArchiMate elements are used per layer (Business, Application, Technology, etc.).
- Valid and semantically correct relationships are used between elements.
- Each view clearly addresses a specific stakeholder concern.
- Diagrams are visually clean, structured, and layered consistently with reference to the SebTreb rules.
- A cross-layer view has been included (e.g., Business Service to Application Service).
- The overall architecture aligns with the goals of the case study (digital claims modernization).

Expected views

The table below highlights the expected views per architecture role/focus.

Views for EA	Views for SA
Goals and Objectives view	Service Contextual view
Business Contextual view	Service Conceptual view
Business Conceptual view	Logical Data Model view
Business Information Model view	Logical Component view
Logical Services view	Deployment or Infrastructure view

Modeling Template Suggestions

The following modeling templates are suggested to help structure your ArchiMate views. You may adjust them as needed for your use case.

1. Motivation View – Stakeholder, Driver, Goal, Outcome, Assessment, Requirement
2. Capability Map – Capability, Resource, Course of Action
3. Business Layer View – Business Actor, Business Role, Business Process, Business Service
4. Application Collaboration View – Application Component, Application Interface, Application Service
5. Technology Infrastructure View – Node, Device, System Software, Technology Interface
6. Cross-Layer View – Business Service to Application to Infrastructure

Assessment Submission Approach

When you complete the assessment, export your model in BizzDesign in **open exchange XML standard** and email to <<***email TBC***>>. Also send a message on the <<***teams channel***>> notifying of your assessment completion.
